Pre-Trained model:

- Pre-trained *models* are neural networks trained on large datasets before being fine-tuned for specific tasks.
- It helps us to achieve high accuracy with less data
- By using pre-trained model, we can save time and computational resources

Pre-trained model for image classification

- ResNet(Residual Network)
- Inception(GoogleNet)
- VGG(Visual Geometry Group)
- EfficientNet
- DenseNet(Dense Convolutional Network)
- MobileNet
- NASNet(Neural Architecture Search Network)
- Xception(Extreme Inception)
- AlexNet
- Vision Transformers(ViT)

ResNet(Residual Network):

- When neural networks get very deep (many layers), they can become harder to train. Instead of improving, they sometimes perform worse because information gets lost or distorted as it passes through so many layers.
- ResNet introduces *shortcut connections* (also called *skip connections*). These shortcuts allow the network to "skip" certain layers and pass information directly forward.
- By skipping layers, the network can focus on learning only the *differences* (residuals) between inputs and outputs, rather than relearning everything from scratch.
- This makes training faster, more stable, and allows for much deeper networks (like 152 layers) without the usual problems.

Variants: ResNet-50, ResNet-101, ResNet-152.

Inception(GoogleNet):

- GoogLeNet doesn't just stack convolution layers one after another. Instead, it groups them into **Inception modules**.
- Each convolutional layer scans the image for features like edges, textures, or shapes.

  Inside an Inception module:

  - **1×1 convolutions**: Reduce the number of channels (like compressing data) and also capture fine details.
  - **3×3 convolutions**: Look for medium-sized patterns (like corners or small shapes).
  - **5×5 convolutions**: Capture larger patterns (like whole objects).
  - **Pooling + 1×1 convolution**: Adds another perspective by summarizing regions and then refining them
- All these convolutional outputs are **concatenated (joined together)**, so the network sees the image at multiple scales at once.

  Variants: Inception v3, Inception v4, Inception-ResNet

VGG(Visual Geometry Group)

- VGG is just a **stack of many small convolutional layers (3×3)** with pooling in between, followed by fully connected layers at the end
- It uses **very small filters (3×3 convolutions)**, but stacks many of them in a row.
- By stacking, the network can capture more complex patterns step by step.

  Variants: VGG-16, VGG-19.

EfficientNet:

- Instead of just making networks bigger or deeper, it uses a smart method called *compound scaling* to balance width, depth, and resolution, so the model learns better without wasting resources.
- EfficientNet models often outperform older networks like ResNet or VGG while using less memory and computation.

  Variants: EfficientNet-B0 to EfficientNet-B7.

DenseNet (Dense Convolutional Network):

- In DenseNet, each layer shares its output with all later layers.
- This means later layers don't have to "relearn" features — they can reuse what earlier layers already found.
- **Efficient learning**: No wasted effort; features are reused.
- **Fewer parameters**: Because it doesn't need to relearn the same patterns, DenseNet is smaller and lighter than many other deep networks.
- **Better gradient flow**: During training, information flows more smoothly backward through the network, making it easier to train deep models.

  Variants: DenseNet-121, DenseNet-169, DenseNet-201.

NASNet (Neural Architecture Search Network):

- AI tried out many possible network designs, tested them and learned which ones worked best, and improved its choices over time.
- The final design discovered was called NASNet.

  Variants: NASNet-A, NASNet-B, NASNet-C.

Xception (Extreme Inception):

- Xception is an extension of the Inception architecture with depthwise separable convolutions.
- Xception splits this into two steps:
  - Depthwise convolution → Each filter looks at one channel separately (like scanning red, green, and blue layers individually).
  - Pointwise convolution (1×1) → Combines all those channels back together.

AlexNet :

- One of the first models trained on GPUs, which made it possible to handle such a large network.
- **Layers**: It has 8 layers — 5 convolutional layers and 3 fully connected layers.
- **ReLU activation**: Uses a fast activation function (ReLU) that helped speed up training.
- **Dropout**: Introduced dropout to reduce overfitting (preventing the network from memorizing instead of learning).

Vision Transformers (ViT)

- They apply the **Transformer architecture** (originally designed for language tasks like translation) to images.
- Instead of scanning with convolutions (like CNNs), they treat an image as a sequence of patches — kind of like breaking a picture into puzzle pieces.
- **Split the image into patches** (like cutting a photo into small squares).
- **Flatten each patch** and turn it into a vector (like turning each puzzle piece into a list of numbers).
- **Add positional encoding** so the model knows where each patch belongs in the image.
- **Feed patches into a Transformer** — the same mechanism used in language models, which uses *attention* to figure out which parts of the image are important.
- **Classify the image** based on the combined information.

[Top Pre-Trained Models for Image Classification - GeeksforGeeks]

Pre-trained model for object Detection:

- YOLOV7

- EfficientDet
- RetinaNet
- Faster R-CNN
- Mask R-CNN
- CenterNet
- DETR(Detection Transformer)
- Cascade R-CNN
- SSD(Single Shot Multibox Detector)
- FCOS(Fully Convolutional One Stage Object Detection)

YOLOV7:

- It belongs to the **YOLO (You Only Look Once)** family of models, which are designed for **real-time object detection** — spotting and labelling things in images or videos very quickly.
- **YOLO principle**: Instead of scanning an image piece by piece, YOLO looks at the whole image in one go and predicts where objects are.
- **YOLOv7 improvements**:
  - Faster and more accurate than earlier YOLO versions.
  - Uses clever tricks like **extended efficient layers** and **better training strategies**.
  - Can detect multiple objects in real time (cars, people, animals, etc.) with high precision.

EfficientNet:

- It builds on EfficientNet as its backbone and introduces innovations like BiFPN (Bi-directional Feature Pyramid Network) and compound scaling to balance speed and accuracy.
- Key Features of EfficientDet
  - **Backbone:** Uses EfficientNet, a family of CNNs optimized for efficiency and accuracy.
  - **BiFPN (Bi-directional Feature Pyramid Network):** Allows fast and easy fusion of features at multiple scales, improving detection of objects of varying sizes.

- **Compound Scaling:** Uniformly scales resolution, depth, and width across backbone, feature network, and prediction layers for balanced performance.
- **Performance:** Achieves state-of-the-art accuracy while being lightweight compared to other detectors like Faster R-CNN or YOLO.

RetinaNet:

- RetinaNet is a deep learning model for object detection that works in a single step ("one-stage") and is known for its accuracy in finding both large and small objects.
- It combines:
  - **Feature Pyramid Network (FPN):** Helps detect objects at different sizes and scales.
  - **Focal Loss:** A new loss function that reduces the imbalance between easy and hard examples during training.

Faster R-CNN:

- Faster R-CNN is a two-stage deep learning model for object detection that first proposes possible regions where objects might be and then classifies them.
- **Two-stage detection:**
  - **Region Proposal Network (RPN):** Suggests candidate regions that may contain objects.
  - **Classification + Bounding Box Regression:** Determines what the object is and refines its location.
  - **High accuracy:** Performs exceptionally well on benchmarks like COCO and PASCAL VOC.
  - **Versatility:** Can detect multiple objects of different sizes in one image.
  - **Backbones:** Often uses ResNet, VGG, or other CNNs to extract features.

Mask R-CNN:

- Mask R-CNN is a computer vision model that not only finds objects in an image but also draws their exact outlines, pixel by pixel.
- Step-by-step process of Mask R-CNN

    - Find regions of interest (RoI); The model guesses which parts of the image might contain objects
    - Classify objects: It decides what each region contains (dog, car, person)
    - Draw bounding boxes: It refines the rectangle around each object
    - Generate masks: For each object, it highlights the exact pixel it belongs to, like tracing the outline

CenterNet:

- CenterNet is an object detection model that finds objects by predicting their *center points* directly, instead of relying on pre-defined anchor boxes
- **Anchor-free approach:** Traditional models like Faster R-CNN or YOLO use *anchors* (pre-set boxes of different sizes/shapes) to guess where objects might be. Center Net skips this step, making detection simpler and faster.
- **Objects as points:** Each object is represented by its **center point** on a heatmap. The model predicts:
    - **Center location** (where the object is in the image).
    - **Size (width & height)** of the bounding box.
    - **Other attributes** like pose or key points (for tasks like human pose estimation).
- **Heatmap prediction:** The network outputs a heatmap where bright spots (high confidence) indicate likely object centres. From these, bounding boxes are drawn.

DETR(Detection Transformer):

- DETR (Detection Transformer) is a breakthrough object detection model that uses Transformers to predict objects directly in a single step.

- Unlike older models, it doesn't rely on anchor boxes or extra steps like non-maximum suppression.
- Working process of DETR:
    - Image features are extracted using a CNN backbone.
    - A Transformer encoder processes these features globally (so every part of the image can see every other part).
    - A Transformer decoder uses *object queries* to predict a fixed set of objects.
    - Each query outputs either:
      An object class + bounding box, or no object

Cascade R-CNN:

- Cascade R-CNN is an advanced object detection model that improves accuracy by using a multi-stage pipeline, that progressively refines bounding boxes and classifications with stricter criteria.
- In simple words, instead of detecting objects in one go, it uses a cascade of detectors to polish results step by step.
- Working process of Cascade R-CNN,
    - **Built on Faster R-CNN:** It kicks off with region proposals created by a Region Proposal Network (RPN).
    - **Multi-stage refinement:**
        - Stage 1: Detects objects with a low Intersection over Union (IoU) threshold (looser boxes, high recall).
        - Stage 2+: Each stage uses a higher IoU threshold, refining bounding boxes and filtering out false positives.
    - **Cascade bounding box regression:** Each stage improves the localisation of objects by adjusting bounding boxes more precisely.
    - **Cascade classification:** Each stage also reclassifies proposals with stricter confidence, reducing noisy detections.

SSD(Single Shot Multibox Detector)

- The Single Shot Multibox Detector (SSD) is a deep learning–based object detection algorithm that performs detection in a single forward pass, making it much faster than two-stage methods like Faster R-CNN.

- It eliminates the need for region proposals and achieves real-time performance while maintaining competitive accuracy.
- Working process of SSD:
  - **Object Detection in One Pass**: Unlike R-CNN or Faster R-CNN, SSD does not generate region proposals. Instead, it predicts bounding boxes and class scores directly from feature maps in a single step.
  - **Default Boxes (Anchor Boxes)**: At each feature map location, SSD defines multiple default boxes of different scales and aspect ratios. These serve as candidate bounding boxes.
  - **Multi-Scale Feature Maps**: SSD uses feature maps from different layers of the network (e.g., VGG16 backbone + extra convolutional layers) to detect objects of varying sizes.
  - **Predictions**: For each default box, SSD predicts:
    - **Class scores** (probability of object categories)
    - **Bounding box offsets** (adjustments to better fit the object)

FCOS(Fully Convolutional One Stage Object Detection)

FCOS (Fully Convolutional One-Stage Object Detection) is a modern anchor-free object detection framework that predicts objects directly at the pixel level, making detection simpler, faster, and more accurate compared to anchor-based methods like Faster R-CNN, SSD, or YOLO.

1. Feature Extraction: Uses a backbone CNN (e.g., ResNet, ResNeXt) to extract feature maps.
2. Per-pixel Classification: Each pixel predicts object category scores.
3. Bounding Box Regression: Each pixel regresses a 4D vector (l, t, r, b)representing distances to the bounding box edges.
4. Center-ness Prediction: Helps suppress low-quality detections by estimating how close a pixel is to the object center.
5. Post-processing: Non-Maximum Suppression (NMS) removes duplicate detections.

Object Detection Models - GeeksforGeeks

PRE-TRAINED MODEL FOR FACE RECOGNITION:

FACENET:

- FaceNet, in simple words, is a **computer program made by Google that can recognize faces by turning them into numbers**.
- Working process of Facenet:
  - **Input**: You give it a picture of a face.
  - **Embedding**: It converts that face into a unique set of numbers (a vector). Think of it like a "digital fingerprint" of the face.
  - **Comparison**: To check if two faces are the same person, it compares their number sets. If the numbers are very close, it's the same person; if they're far apart, it's different people.

PRE-TRAINED MODELS FOR TEXT:

- BERT
- GPT-2
- ELMo
- Transformer-XL
- RoBERTa

Top 5 PreTrained Models in Natural Language Processing (NLP) - GeeksforGeeks